GHENA HANI

Applied AI Engineer | NLP, Deep Learning & LLM Systems

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EXECUTIVE SUMMARY

Applied AI Engineer & M.Sc. Candidate in Artificial Intelligence at USTHB (Sep 2025 - Expected June 2027), Licence in Computer Science graduate from USTHB (2022 - 2025), with AI research appointments at CERIST (2024 - 2025) and commercial full-stack software delivery (Jan 2024 - Present). Proven track record designing evaluated deep learning architectures (AsthmaExpertUltra Multi-Task Transformer, MuCAT 6-language Transformer, MIT-BIH v7 INT8 TFLite ECG classifier), real-time IoT clinical platforms (MedGuardAI), hybrid RAG pipelines, and 6 deployed commercial client systems.

PROFESSIONAL & RESEARCH EXPERIENCE

AI Engineering Research Intern - CERIST (UbiSys Team DTISI), Algiers (2024 - 2025)

- Designed AsthmaExpertUltra: 4-layer Transformer Encoder predicting crisis severity (76.0% acc, 92.3% severe recall) and lead-time proximity (69.0% acc) from 8 multimodal IoT streams, with Dempster-Shafer fusion & Android PyTorch Lite deployment (<1.0s).
- Designed MuCAT (Multilingual Uncertainty-Calibrated Attention Transformer) in CERIST UbiSys (DTISI): mDeBERTa-v3-base + Hierarchical Attention Pooling (HAP) + FiLM Language Gating + Evidential Dirichlet Head (u=K/S) across 6 languages (MSA, Algerian Darija, Kabyle, Chaoui, FR, EN), reaching 98.20% test accuracy.
- Engineered AI ECG Arrhythmia Classification (v7) on the strict inter-patient MIT-BIH DS1/DS2 benchmark (90.13% 5-fold accuracy, 155.4 KB INT8 TFLite, 0.32 ms/beat latency).

Independent Full-Stack & Applied AI Engineer - Algiers, Algeria (Jan 2024 - Present)

- Architected and delivered 6 commercial production systems: Tadjmeel Clinica Algiers (Public Web + Doctor & Receptionist Electron Desktop Apps), Enterprise DZD Business ERP & G50 Fiscal Suite, ALLURE HOMME, GK STORE, CASUAL 29, and AURA Retail OS.
- Built and deployed 11+ interactive AI web applications spanning Speech-to-Text ASR & Clinical SOAP Summarization, Hybrid Dense+BM25 RAG, Multi-Agent ReAct Orchestration, and LLM Safety Evaluation.

KEY DEPLOYED SYSTEMS & RESEARCH

- AsthmaExpertUltra (CERIST AI Research): Multi-Task Transformer for asthma crisis severity & lead-time with Android PyTorch Lite.
- AI ECG Arrhythmia Classification (v7): 1D-CNN + 2x Multi-Head Transformer + 8 RR features on MIT-BIH DS1/DS2.
- MuCAT Multilingual Transformer: mDeBERTa-v3 + HAP + FiLM Gating + Dirichlet Uncertainty (98.20% Test Acc, 6 languages).
- MedGuardAI (USTHB Capstone Distinction): Real-time IoT ECG/Glucose sensor streaming, AI anomaly alerts & WebRTC telemedicine.
- Commercial Production Suite: Tadjmeel Clinica Algiers, Enterprise DZD G50 ERP, ALLURE HOMME, GK STORE & CASUAL 29.

EDUCATION

M.Sc. Candidate in Artificial Intelligence - USTHB, Algiers (Sep 2025 - June 2027 Expected)

B.Sc. (Licence) in Computer Science - USTHB, Algiers (Sep 2022 - June 2025 Graduated - Capstone: MedGuardAI)

Baccalaureat in Exact Sciences - Algiers (2022)

TECHNICAL COMPETENCIES & LANGUAGES